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(4)

Measures of Central Tendency

Measures of Central Tendency (The Median)

In the case of GFD, we first find the median position and identify the class where the median position is found. This class is also known as the *median class*. Then to find the value of the median, we use the following formula which we call the *interpolation method*.

Measures of Central Tendency (The Median)

$$\tilde{x}_{val} = LCB + \left[\frac{\frac{N}{2} - \sum F_i}{f_{median}} \right] \times w, \text{ Where}$$

LCB – Lower class boundary of the median class.

N – Number of observations

$\sum F_i$ – Cumulative frequency of a class just before the median class.

f_{median} – The frequency of the median class.

w – The class width of the median class.

Measures of Central Tendency (The Median)

Example 1:- The following GFD refers to the weight (to the nearest Kg) of a sample of students at HiLCoE. Find the median weight and interpret your result.

Weight (in Kg)	Number of students	Cum. Frequency (Less than type)
35 - 43	8	
44 - 52	11	
53 - 61	16	
62 - 70	19	
71 - 79	14	
80 - 88	9	

Measures of Central Tendency (The Median)

Weight (in Kg)	Number of students	Cum. Frequency (Less than type)
35 - 43	8	8
44 - 52	11	19
53 - 61	16	35
62 - 70	19	54
71 - 79	14	68
80 - 88	9	77

Median Class

$$\text{Here } \tilde{x}_{pos} = \left(\frac{N+1}{2} \right)^{th} = \left(\frac{77+1}{2} \right)^{th} = \left(\frac{78}{2} \right)^{th} = (39)^{th}$$

As we can see the 39th student is in the 4th class.

Measures of Central Tendency (The Median)

$$\begin{aligned}\tilde{x}_{val} &= \text{LCB} + \left[\frac{\frac{N}{2} - \sum F_i}{f_{median}} \right] \times w = 61.5 + \left[\frac{\frac{78}{2} - 35}{19} \right] \times 9 \\ &= 61.5 + \left[\frac{39 - 35}{19} \right] \times 9 = 61.5 + \left[\frac{4}{19} \right] \times 9 = 61.5 + \left[\frac{36}{19} \right] \\ &= 61.5 + 1.89 = 63.4\end{aligned}$$

Interpretation:- About 50% of the students in the sample weight is less than (or equal to) 63.4 Kg [or more than (or equal to) 63.4 Kg.]

Measures of Central Tendency (The Median)

Advantages of Median:

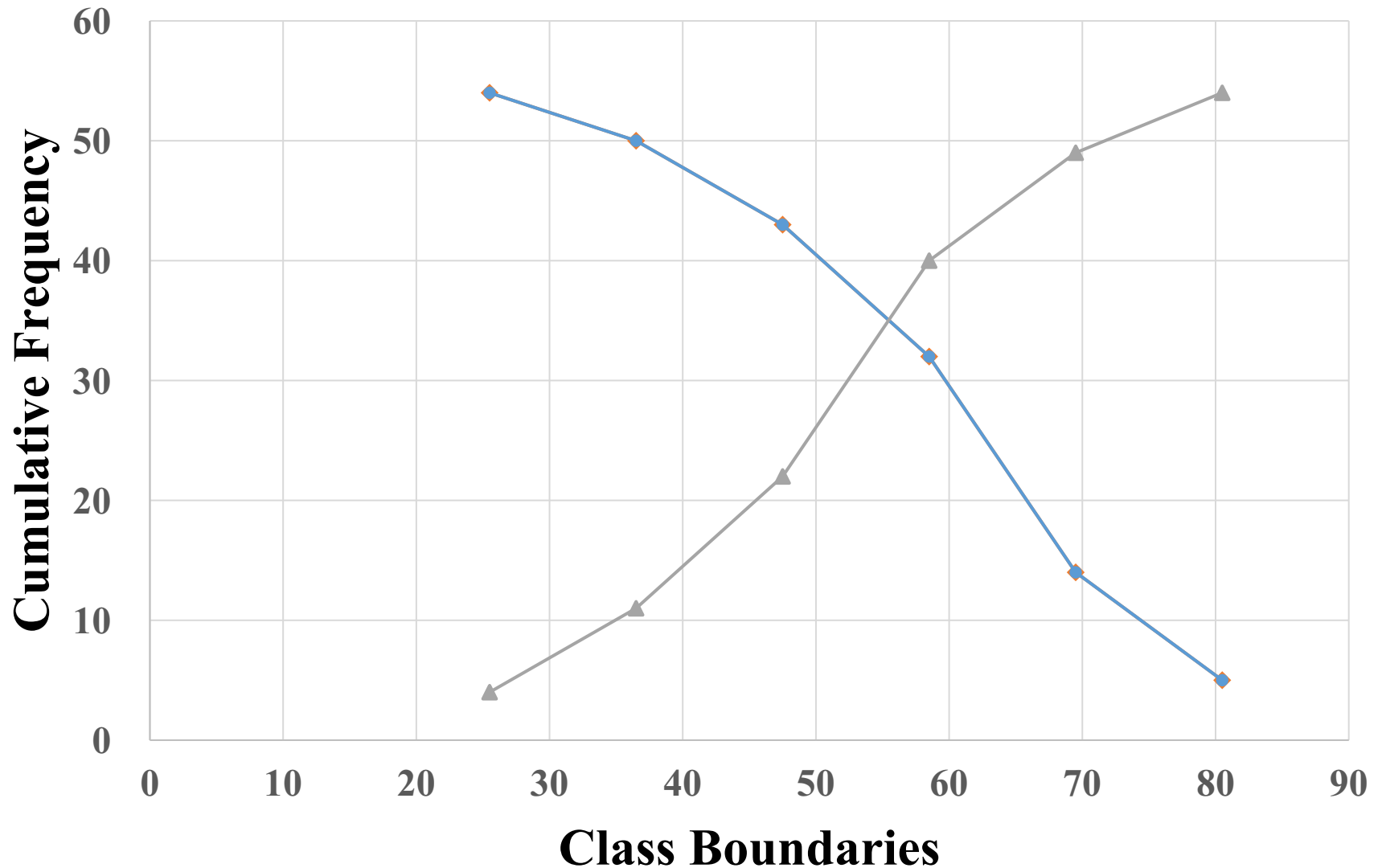
- It is very simple to understand and easy to calculate.

In some cases it is obtained simply by inspection.

- Median lies at the middle part of the series and hence it is not affected by the extreme values.
- In grouped frequency distribution it can be graphically located by drawing ogives.

Measures of Central Tendency (The Median)

Ogives presenting the weight (to the nearest Kg) of a sample of HiLCoE students registered for the online class in Spring 2020



Measures of Central Tendency (The Median)

It is a special average used in qualitative phenomena like intelligence or difficulty or simplicity of an examination which are not quantified but ranks are given. Thus we can locate the person whose intelligence is the average or the weight of an exam giving ranks as very difficult, difficult, fair, simple, very simple and identify the average.

VD, VD, VD, D, D, D . . . , . D, D, F, F, F, . . . , S, S, S, VS, VS , VS



$\tilde{X}_{\text{Position}}$

About 50% of students evaluated the exam as difficult and very difficult

Measures of Central Tendency (The Median)

Disadvantages of Median :

❑ In simple series, the item values have to be arranged.

If the series contains large number of items,
then the process becomes tedious.

❑ It is a less representative average because it does
not depend on all the items in the series.

Measures of Central Tendency (The Median)

- It is not capable of further algebraic treatment.

For example, we can not find a combined median of two or more groups if the median of different groups are given.

- It is affected more by sampling fluctuations than the mean as it is concerned with only one item i.e. the middle item.

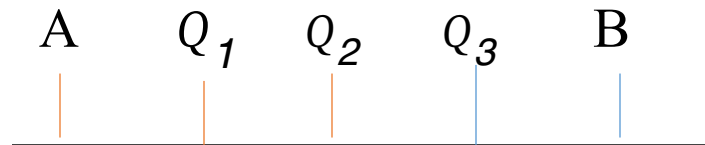
Measures of Central Tendency (The Median)

- It is not rigidly defined. In simple series having even number of items, median cannot be exactly found. Moreover, the interpolation formula applied in the continuous series is based on the unrealistic assumption that the frequency of the median class is evenly spread over the magnitude of the class interval of the median group.

Quartiles, Deciles & Percentiles

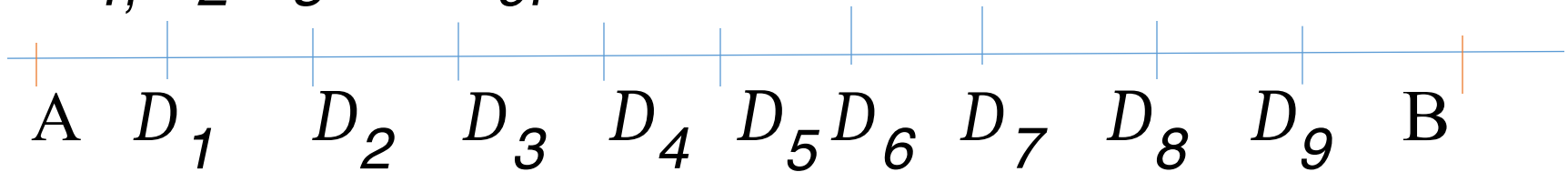
MCT (Quartiles, Deciles and Percentiles)

By extending the idea of the median, we can think of those values which divide the data set in to four equal parts. These values, denoted by Q_1 , Q_2 , and Q_3 are called first (lower), second (middle) and third (upper quartiles respectively, where Q_2 being equal to the median.



MCT (Quartiles, Deciles and Percentiles)

- Similarly the values which divide the data set in to ten equal parts are called deciles and are denoted by $D_1, D_2, D_3 \dots D_9$.



- While the values which divide the data set in to 100 equal parts are called percentiles and are denoted by $P_1, P_2, P_3 \dots P_{99}$.

MCT (Quartiles, Deciles and Percentiles)

- Again note that the 2nd quartile, the 5th decile and the 50th percentile correspond to the median and the 25th and the 75th percentiles correspond to the 1st and 3rd quartiles respectively.

MCT (Quartiles, Deciles and Percentiles)

- To find the positions of quartiles, deciles and percentiles for the given Grouped Frequency Distribution, we use the following formulas and to find the respective values we use the interpolation method as we used in the case of median for the GFD.

MCT (Quartiles, Deciles and Percentiles)

$$Q_i = \left[\frac{i \times N}{4} \right]_t h$$

$$D_i = \left[\frac{i \times N}{10} \right]_t h$$

$$P_i = \left[\frac{i \times N}{100} \right]_t h$$

MCT (Quartiles, Deciles and Percentiles)

Example 1:- The following GFD refers to the weight (to the nearest Kg) of a sample of students at HiLCoE. Find $Q_1, Q_3, D_2, D_3, D_8, P_{24}, P_{39}$ and P_{88} and interpret your result.

Weight (in Kg)	Number of students	Cum. Frequency (Less than type)
35 - 43	8	8
44 - 52	11	19
53 - 61	16	35
62 - 70	19	54
71 - 79	14	68
80 - 88	9	77

MCT (Quartiles, Deciles and Percentiles)

Solution i:- $Q_{1(\text{pos})} = \left[\frac{1 \times 77}{4} \right]^{th} h = \left[\frac{77}{4} \right]^{th} h =$
 $(19.25)^{th} h \approx 19^{th} h$

You can see that the $19^{th} h$ student is in the second class.

Now using interpolation method we get

$$Q_{1(\text{val})} = \text{LCB} + \left[\frac{Q_{1(\text{pos})} - \Sigma F_i}{f_{Q1}} \right] \times w$$

Thus $Q_{1(\text{val})} = 43.5 + \left[\frac{19 - 8}{11} \right] \times 9 = 43.5 + \left[\frac{11}{11} \right] \times 9 = 52.5$

MCT (Quartiles, Deciles and Percentiles)

Interpretation:- About 25% of the students in the sample weight is less than (or equal to) 52.5 Kg or about 75 % of the students in the sample weight is more than (or equal to) 52.5 Kg.

Solution ii:- $Q_{3(\text{pos})} = \left[\frac{3 \times 77}{4} \right]^{th} = \left[\frac{231}{4} \right]^{th} =$
 $(57.75)^{th} \approx 58^{th}$

You can see that the 58^{th} student is in the fifth class.

MCT (Quartiles, Deciles and Percentiles)

Now using interpolation method we get

$$Q_{3(\text{val})} = \text{LCB} + \left[\frac{Q_{3(\text{pos})} - \sum F_i}{f_{Q3}} \right] \times w$$

$$\begin{aligned} \text{Thus } Q_{3(\text{val})} &= 70.5 + \left[\frac{58 - 54}{14} \right] \times 9 = 70.5 + \frac{36}{14} \\ &= 70.5 + 2.57 = 73.07 \approx 73.1 \end{aligned}$$

Interpretation:- *About 75% of the students in the sample weight is less than (or equal to) 73.1 Kg or about 25 % of the students in the sample weight is more than (or equal to) 73.1 Kg.*

MCT (Quartiles, Deciles and Percentiles)

Solution iii:- $D_{2(Pos)} = \left[\frac{2 \times 77}{10} \right]^{th} = 15.4^{th} \approx 15^{th}$

As you can see from the given table, the 15^{th} student is in the second class.

Now using interpolation method we get

$$D_{2(val)} = LCB + \left[\frac{D_{2(Pos)} - \sum F_i}{f_{D2}} \right] \times w$$

$$\begin{aligned} \text{Thus } D_{2(val)} &= 43.5 + \left[\frac{15 - 8}{11} \right] \times 9 \\ &= 43.5 + \frac{63}{11} = 43.5 + 5.73 = 49.23 \end{aligned}$$

MCT (Quartiles, Deciles and Percentiles)

Interpretation:- About 20% of the students in the sample weight is less than (or equal to) 49.23 Kg or about 80 % of the students in the sample weight is more than (or equal to) 49.23 Kg.

Solution iv:- $D_{3(Pos)} = \left[\frac{3 \times 77}{10} \right]^{th} = 23.1^{th} \approx 23^{rd}$

As you can see from the given table, the 23^{rd} student is in the third class.

MCT (Quartiles, Deciles and Percentiles)

Now using interpolation method we get

$$D_{3(val)} = LCB + \left[\frac{D_{3(Pos)} - \sum F_i}{f_{D3}} \right] \times w$$

$$\begin{aligned} \text{Thus } D_{3(val)} &= 52.5 + \left[\frac{23 - 19}{16} \right] \times 9 = 52.5 + \left[\frac{4}{16} \right] \times 9 \\ &= 52.5 + \frac{36}{16} = 52.5 + 2.25 = 54.75 \end{aligned}$$

Interpretation:- About 30% of the students in the sample weight is less than (or equal to) 54.8 Kg or about 70 % of the students in the sample weight is more than (or equal to) 54.8 Kg.

MCT (Quartiles, Deciles and Percentiles)

Solution v:- $D_{8(Pos)} = \left[\frac{8 \times 77}{10} \right]^{th} = 61.6^{th} \approx 62^{nd}$

As you can see from the given table, the 62^{nd} student is in the fifth class. Now using interpolation method we get

$$D_{8(val)} = LCB + \left[\frac{D_{8(Pos)} - \sum F_i}{f_{D8}} \right] \times w$$

$$\begin{aligned} \text{Thus } D_{8(val)} &= 70.5 + \left[\frac{62 - 54}{14} \right] \times 9 = 70.5 + \left[\frac{8}{14} \right] \times 9 \\ &= 70.5 + \frac{72}{14} = 70.5 + 5.14 \approx 75.6 \end{aligned}$$

MCT (Quartiles, Deciles and Percentiles)

Interpretation:- About 80% of the students in the sample weight is less than (or equal to) 75.6 Kg or about 20 % of the students in the sample weight is more than (or equal to) 75.6 Kg.

Solution vi:- $P_{24(Pos)} = \left[\frac{24 \times 77}{100} \right]^{th} = 18.48^{th} \approx 18^{th}$

As you can see from the given table, the 18th student is in the second class. Now using interpolation method we get

MCT (Quartiles, Deciles and Percentiles)

$$P_{24(val)} = LCB + \left[\frac{P_{24(Pos)} - \sum F_i}{f_{P24}} \right] \times w$$

$$\begin{aligned} \text{Thus } P_{24(val)} &= 43.5 + \left[\frac{18 - 8}{11} \right] \times 9 \\ &= 43.5 + \frac{80}{11} = 43.5 + 7.27 \approx 50.8 \end{aligned}$$

Interpretation:- About 24% of the students in the sample weight is less than (or equal to) 50.8 Kg or about 76 % of the students in the sample weight is more than (or equal to) **50.8 Kg.**

MCT (Quartiles, Deciles and Percentiles)

Solution vii:- $P_{39}(\text{Pos}) = \left[\frac{39 \times 77}{100} \right]^{th} h =$
 $30.03^{th} h \approx 30^{th}$

As you can see from the given table, the 30th student is in the third class. Now using interpolation method we get

$$P_{39}(\text{val}) = \text{LCB} + \left[\frac{P_{39}(\text{Pos}) - \sum F_i}{f_{P39}} \right] \times w$$

[30 10]

MCT (Quartiles, Deciles and Percentiles)

Interpretation:- About 39% of the students in the sample weight is less than (or equal to) **61.5** Kg or about 61 % of the students in the sample weight is more than (or equal to) 61.5 Kg.

MCT (Quartiles, Deciles and Percentiles)

Solution viii:- $P_{88(Pos)} = \left[\frac{88 \times 77}{100} \right]^{th} = 67.76^{th} \approx 68^{th}$

As you can see from the given table, the 68th student is in the fifth class. Now using interpolation method we get

$$P_{88(val)} = LCB + \left[\frac{P_{88(Pos)} - \Sigma F_i}{f_{P88}} \right] \times w$$

$$\text{Thus } P_{88(val)} = 70.5 + \left[\frac{68 - 54}{14} \right] \times 9 = 70.5 + 9 \approx 79.5$$

Interpretation:- About 88% of the students in the sample weight is less than (or equal to) 79.5 Kg or about 12 % of the students in the sample weight is more than (or equal to) 79.5 Kg.

Any Questions



Any Questions



Thank
you